Prathamesh Nitin Tanksale

“Only thing certain in life is uncertainty in life.”

Contact details

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BW-310, Beas Hostel, IIT Ropar permanent campus, Rupnagar 140001, Punjab, India.

Teaching Assitance

- Security aware architecture, Memory system and architecture, Computer systems, Digital logic design, Computer architecture, Data structures and Algorithms

Skills

- C, C++, Python
- LaTeX, Shell script
- VS Code, MS Word, Excel, PowerPoint

Simulators

- ChampSim, CACTI
- Ramulator, NVSIM

Personal information

Citizenship: **India**

Languages: **English**, **Hindi**, **Marathi** (native)

Hobbies

- Reading spiritual texts such as the Bhagavad Gita and Śrīmad Bhāgavatam
- Playing chess, Debating, Anchoring, Cooking and exploring new cuisines

Profile

Ph.D. Scholar at IIT Ropar specializing in microarchitectural security and high-performance CPU design. Experienced in developing and validating architectural simulators for performance and energy analysis. Proficient in C/C++ programming with strong expertise in system-level modeling and hardware–software co-design. Research focuses on bridging CPU performance optimization with secure architecture design, with publications in leading venues such as IEEE HOST and DATE.

Research Interests

Security-Oriented: Microarchitectural security, side-channel resilient cache architectures, secure memory hierarchies, and hardware-level defense mechanisms for emerging memory technologies.

Performance-Oriented: High-performance CPU and memory system design, cache optimization and management, scalable architectures for emerging non-volatile memories, and performance–security co-optimization in modern microarchitectures.

Publications

Conference

◇ Prathamesh Nitin Tanksale, Guru Raghava S. Seethiraju, Shirshendu Das, Venkata Kalyan Tavva: $\mathcal{RRR}$: Rethinking Randomized Remapping for High Performance Secured NVM LLC. **HOST 2025**, pp. 381-391. https://doi.org/10.1109/HOST64725.2025.11050039

◇ Aditya S. Gangwar, Prathamesh Nitin Tanksale, Shirshendu Das, Sudeepta Mishra: $\mathcal{F}lush + early\mathcal{R}$ELOAD: Covert Channels Attack on Shared LLC Using MSHR Merging. **DATE 2024** pp. 1-6. https://doi.org/10.23919/DATE58400.2024.10546609

◇ Prathamesh Nitin Tanksale, Kousik Kumar Dutta, Shirshendu Das, Kalyan, Venkata Kalyan Tavva: CAMOUFLAGE: An Efficient Mechanism to Hide Congestion in NVM LLC, Extended Abstract in **HiPC 2023**.

◇ Kousik Kumar Dutta, Prathamesh Nitin Tanksale, Shirshendu Das: A fairness conscious cache replacement policy for last level cache. **DATE 2021**, pp. 695–700 (2021). https://doi.org/10.23919/DATE51398.2021.9474096

Education

Ph.D. in Computer Science and Engineering. *Indian Institute of Technology Ropar* (CGPA 8.4). **2021–ongoing**

◇ Thesis title: *Performance-Centric Redesign of Secure Cache Architectures.*

◇ Courses: Security aware architecture, Memory systems and architecture, Advanced computer architecture, Scheduling in Parallel and Distributed Systems.

M.Tech. in Computer Science and Engineering. *Indian Institute of Technology Ropar* (CGPA 8.6). **2019–2021**

◇ Thesis title: *An Efficient Mechanism to Hide Congestion in NVM LLC.*

◇ Data structures and algorithm, Computer systems, Advanced operating systems.